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Communicating Cascading Natural Hazards through Co-Created Visualization with Youth

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Effective communication of complex geoscience events, such as cascading natural hazards, is challenging, particularly when addressing non-technical audiences. This study explores an approach used to raise awareness among younger generations by co-creating an interactive visualization of cascading hazards associated with landslides. The Göta River valley in Sweden is a region highly susceptible to landslides and was used as a case study for engaging upper secondary school students (ages 17–18) in a series of workshops between 2020 and 2024. The workshops combined lectures on cascading landslide dynamics with participatory activities to elicit students' perceptions, emotional responses, and preferences for communication formats.

The study integrated descriptive scenario-building, interaction design, and scrollytelling techniques to create a digital visualization prototype of potential cascading natural hazards. Students contributed to the development of a descriptive scenario illustrating a plausible cascade chain of events triggered by prolonged precipitation, leading to erosion, landslides, and secondary impacts such as flood waves and upstream and downstream flooding. Insights from the workshops informed the development of a storyboard design and content, emphasizing students' needs for clarity, concise text, and hopeful messaging. The visualization prototype was implemented using a proprietary web design tool and supplemented with AI-generated illustrations to visualize potential cascading natural hazards in the case study area.

Results indicate that co-creation enhanced engagement and comprehension. Students valued interactive scrollytelling and multimedia elements as complements to traditional static risk maps. Survey responses from the final evaluation workshop showed that 86% of participants found the resulting visualization prototype to be an engaging way to learn about cascading natural hazards, while 78% considered the descriptive scenario easy to follow. Students were eager to learn more about several areas including responsibilities, impacts, and actionable solutions. They valued visual clarity and emotional resonance in the prototype. Critiques highlighted the need for more realistic imagery, even more concise text and interactive elements, and suggested future improvements such as the addition of a glossary.

This study demonstrates the potential of participatory visualization tools to complement conventional hazard communication, encouraging inclusivity and resilience by making complex cascading natural hazard processes accessible and compelling for youth audiences.